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On the foundations of Earth foundation models



Xiao Xiang Zhu^{1,2}✉, Zhitong Xiong^{1,4}, Yi Wang^{1,4}, Adam J. Stewart^{1,2,4}, Konrad Heidler^{1,4},
Yuanyuan Wang¹, Zhenghang Yuan¹, Thomas Dujardin¹, Qingsong Xu^{1,4} & Yilei Shi³

Foundation models have enormous potential in advancing Earth sciences. However, current approaches may not be optimal, as they focus only on a few basic features of a desirable Earth foundation model. Crafting the ideal Earth foundation model, we define eleven features which would allow such a foundation model to be beneficial for any geoscientific downstream application in an environmental- and human-centric manner. We further shed light on the way forward to achieve the ideal model and to evaluate Earth foundation models. What comes after foundation models? Energy efficient adaptation, adversarial defenses, and interpretability are among the emerging directions.

Advances in deep neural networks have revolutionized the world in many domains, including Earth and environmental science. In the past few years, a general trend towards larger datasets (with millions to billions of data points¹) and larger model architectures (with billions to trillions of parameters²) has emerged, with a paradigm shift from single-purpose supervised training to general-purpose self-supervised pre-training. This has been led by the rise in popularity of large language models (LLMs) and vision-language models (VLMs) like CLIP³, LLaMA⁴, GPT⁵, and SAM⁶, spawning billion dollar companies and revolutionizing many aspects of daily life. The term “foundation model” (FM), popularized by Bommasani et al.⁷, encapsulates this emerging model class: “any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks.” FMs allow for emergent capabilities that surpass traditional task-specific models, enabling versatile systems that can be adapted to diverse tasks across major domains—including language, vision, and robotics—which together support the full “perception–reasoning–action” pipeline, with applications across a broad range of fields such as healthcare, law, and education. Their development is driven by the efficiency of training once and adapting to many tasks via fine-tuning or prompting, thereby reducing the need for task-specific data and compute⁷.

While the FM revolution has brought great success in natural language and vision, Earth science presents a distinct and urgent frontier. As our home, the Earth’s surface and atmosphere include the majority of processes that directly impact and are impacted by human activity, which are monitored by vast data from remote sensing satellites, weather stations, and many other sensors. Compared to traditional numerical or machine learning models that are limited by computation at inference, scalability, label scarcity, or transferability, FMs offer a promising pathway forward by

learning generalizable feature representations from large-scale unlabeled data. Crucially, they can support a scalable AI infrastructure for Earth monitoring, forecasting, and decision-making in this complex planetary system towards a sustainable future.

This paper presents a comprehensive framework for defining, developing, and evaluating Earth FMs. We identify eleven critical features that constitute an ideal FM for geoscientific applications, including capabilities such as geolocation and scale awareness, multisensor integration, carbon-efficient operation, and physical consistency. We assess the current progress of Earth FMs against these criteria, highlight key gaps and limitations, and propose a path towards the ideal FM and standardized evaluation benchmarks. What comes after the ideal Earth FM? We identify eight most relevant follow-up research questions such as efficient adaptation and continual updating. Ultimately, this work aims to accelerate the convergence of AI and geoscience in the era of big data, laying the foundation for next-generation tools that are both powerful and sustainable.

The potential of Earth foundation models

Earth science has inevitably entered the era of big data. This data, largely unlabeled and vast in volume, can offer novel insights into our planet’s systems. Earth observation (EO) data-informed modeling offers the potential for higher-resolution and less-biased future projections. However, the sheer volume and redundancy of this information calls for more sophisticated analytical frameworks⁸. FMs are proving to be a glimmer of hope in this context. These EO FMs support learning generalizable, transferable representations that enable adaptation to a wide array of downstream tasks, such as land use/land cover classification, object detection, change detection, visual question answering, and multimodal data fusion⁹. This section delves into the critical reasons underpinning the need for FMs

¹Chair of Data Science in Earth Observation, Technical University of Munich, Munich, Germany. ²Munich Center for Machine Learning, Munich, Germany. ³School of Engineering and Design, Technical University of Munich, Munich, Germany. ⁴These authors contributed equally: Zhitong Xiong, Yi Wang, Adam J. Stewart, Konrad Heidler, Qingsong Xu. ✉e-mail: xiaoxiang.zhu@tum.de

in the Earth sciences, illustrating their potential to revolutionize our understanding of the Earth system towards its sustainable management.

- **Unlocking the value of big EO and climate data:** The primary allure of FMs lies in their ability to harness vast, often unlabeled, reservoirs of EO and climate data by means of learning general representations from them⁷. Traditional models falter under the weight of this information overload, struggling to process and extract meaningful insights without extensive, manually labeled datasets. FMs, on the other hand, can learn task-agnostic general feature representations from massive amounts of unlabeled data. This capability not only improves our understanding of Earth's systems, but also accelerates the pace of discovery from the learned general features.
- **Enhancing label efficiency/boosting performance in downstream tasks:** FMs' ability to learn general representations from vast unlabeled EO and climate data significantly improves label efficiency in downstream tasks. By leveraging self-supervised learning, FMs require only small, task-specific labeled datasets for fine-tuning¹⁰. This significantly accelerates AI applications in EO and enhances model accuracy in various downstream tasks.
- **Reducing carbon footprints:** Traditional methods train models from scratch for each task, consuming significant computational and energy resources¹¹. FMs offer a sustainable alternative by learning general feature representations from big Earth data and only require fine-tuning for new tasks. This approach significantly reduces the computational demand and, consequently, the environmental impact of AI model training in the Earth and climate sciences. FMs also offer the ability to replace or supplement existing physics-driven numerical modeling, often requiring orders of magnitude less energy to run¹².
- **Bridging EO and climate science:** Existing physics-based weather and climate models suffer from imperfect parameterizations of high-resolution microphysics, including clouds and aerosols, ocean eddies, and interactions between the atmosphere and land surface¹³. While deep learning is already being used to downscale and correct for biases in weather and climate models¹⁴, it also shows promise for directly replacing these parameterizations where the physics is either unknown or at a resolution that cannot be simulated at a global scale. In particular, the higher resolution of EO data can help bridge this gap, e.g., by improving estimates of terrestrial carbon fluxes, modeling the relationship between wildfires and aerosols, and mapping sea surface temperature at higher fidelity¹⁵. Earth FMs, e.g., as sketched in Fig. 1, facilitate this integration by providing a framework through which EO data can directly inform and enhance climate models. This is essential

for improving the resolution and prediction accuracy of climate models and reducing their biases and uncertainty, and thus leads to more effective and sustainable climate change mitigation and adaptation strategies.

- **Improving Earth system modeling:** The study of climate change trends and the prediction of future weather and climate scenarios have traditionally relied on general circulation models and numerical weather prediction models. However, the complexity and scale of Earth system modeling pose challenges to computational capacity and efficiency with these conventional methods. Traditional numerical models lack the resolution to accurately model sub-grid scale microphysics, instead relying on approximations of these physical phenomena. FMs offer scale invariance and efficient feature representations, presenting an opportunity for a unified approach to Earth system modeling¹⁶. Furthermore, the complexities and uncertainties of Earth system modeling require an interdisciplinary methodology to advance scientific knowledge discovery. Generative FMs show promise in analyzing diverse data modalities, including images and sequences, to facilitate tasks such as physical law discovery^{2,17}.

To this end, advances in Earth FMs would lead to a paradigm shift in how we approach the vast and complex observational and simulated data characterizing our planet. By unlocking the value of big EO and climate data, enhancing label efficiency, reducing carbon footprints, and bridging the gap between EO and climate science, FMs pave the way for a more sustainable, efficient, and comprehensive understanding of the Earth system. Their potential to revolutionize Earth science is immense, particularly when it comes to informing critical decisions in the face of climate change, e.g., by improving modeling of mitigation and adaptation.

Many of the potentials described in this section are closely tied to the core features we believe an ideal Earth FM should possess. These features are further elaborated in "Crafting the ideal Earth foundation model", where we outline the essential characteristics needed to meet the challenges of Earth modeling.

Big (Earth) data

The development of Earth FMs relies on vast and varied data for self-supervised pre-training. Below, we delve into the most important data sources for both training and evaluation.

- **EO data archive:** To date, there are more than 1000 active remote sensing satellites in space¹⁸. Various EO archives provide a huge volume of data sources for building EO FMs. As Fig. 2a shows, the data

Fig. 1 | The scheme of an Earth foundation model.

An Earth foundation model (FM) should be trained on common data modalities, including imagery (radar, optical), non-Euclidean data (point clouds, text), and meteorological data. It should also provide consistency with physical laws. The FM is task-agnostic, exemplified by five possible downstream tasks. Because of the large difference of the characteristics among Earth observation (EO), weather, and climate data, the resulting FM may consist of a pool of expert models, where feedback loops exist between each of them. Data assimilation may be required, and large language models (LLMs) may be used to provide reasoning capability and natural language interfaces. (For attribution of figure elements, please see Supplementary Note 3).

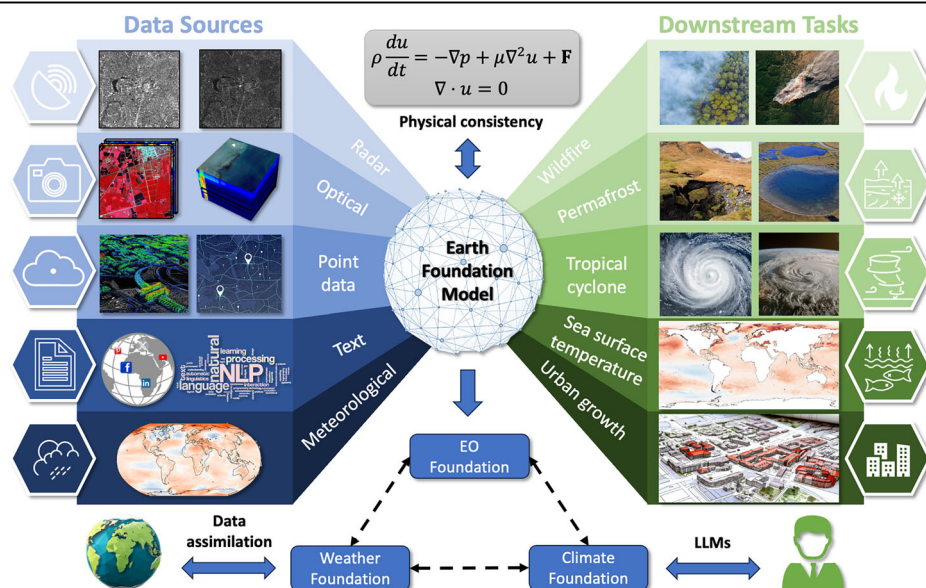
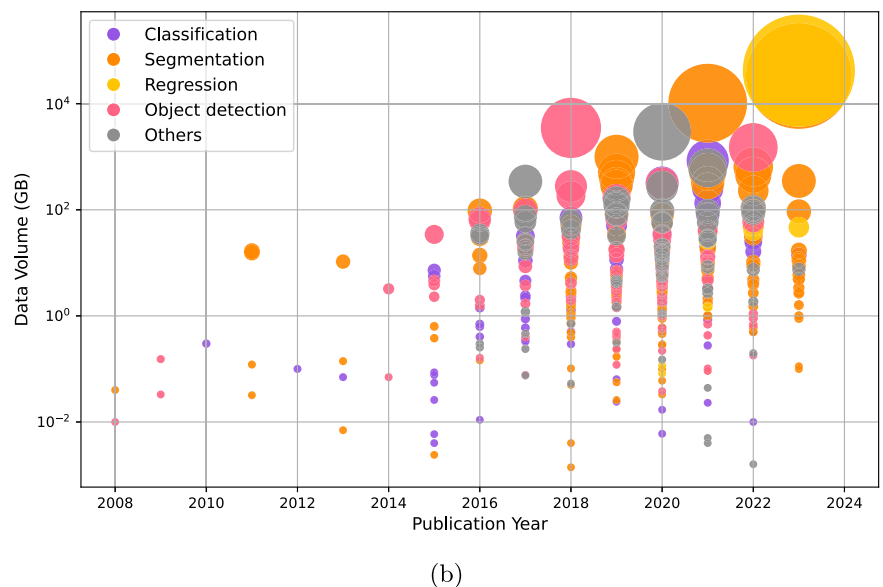
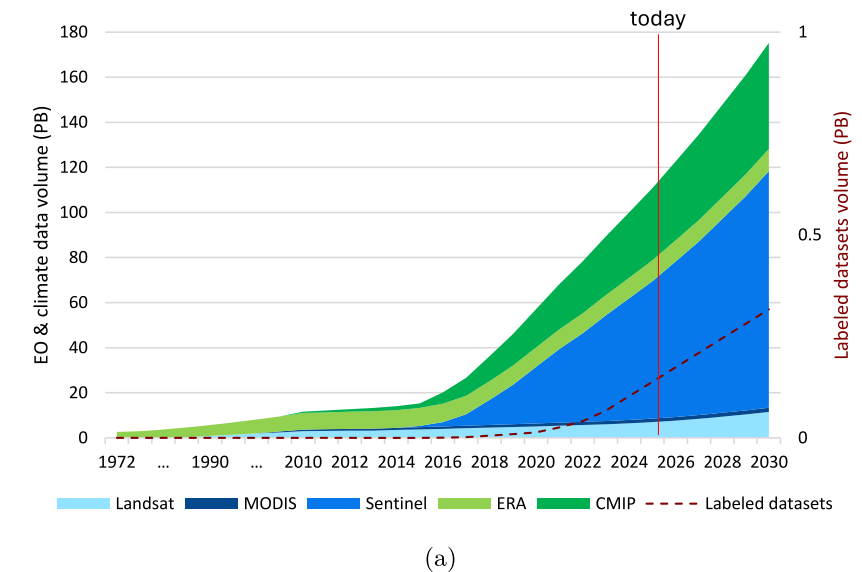


Fig. 2 | Volume of Earth observation and climate data archives and curated benchmark datasets.

a Historical, current, and estimated future volume of open Earth observation (EO) and climate data archives in petabytes (PB)^{29,146–151}. Labeled dataset volumes are shown by a dashed line and use a different y-axis scale. Note that the volume of labeled data is less than 0.1% of the total volume of unlabeled data. **b** Evolution of labeled EO datasets over the last 16 years²⁹. Each bubble represents a single published dataset. Bubble size and y-axis height indicate the volume of labeled data in each dataset in gigabytes (GB). Bubble color denotes the type of dataset. More details on figure creation can be found in Supplementary Note 1.



volume of EO missions whose data is open and free drastically increased over the past decade, surging to nearly 100 PB today, not including commercial data with comparable volume^{19,20}. These archives provide a continuous, high-quality, and open stream of satellite imagery, capturing a wide range of Earth's features at various spatial, spectral, and temporal resolutions. For example, the Sentinel series²¹ includes multiple satellites with different sensors designed to monitor the land, ocean, and atmosphere, and the Landsat series²² commands attention for its extensive temporal reach, providing an invaluable long-term perspective on Earth's surface changes.

- **Weather observations and climate simulations:** In meteorology, various weather and atmospheric science agencies, including ECMWF and NOAA, provide accurate and comprehensive datasets for understanding the Earth's atmosphere. Reanalysis products such as ERA5²³, JRA-55²⁴, NCEP²⁵, and HRRR²⁶ are already widely used for FM pre-training. High-resolution, historical, and near-real-time data on atmospheric conditions make it possible to build large-scale weather nowcasting and medium-range forecasting systems. Furthermore, climate models like CMIP²⁷ and MERRA-2²⁸ offer simulations of past, present, and future climate scenarios built on strong physical

understandings of the climate system that are valuable to climate FMs. Figure 2a highlights a number of these datasets.

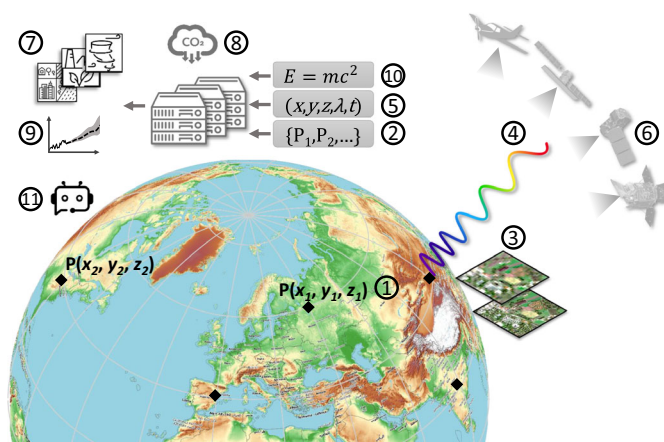
- **Curated ML-ready datasets:** The rise of machine learning (ML) in the past decade has driven the EO community to curate and publish over 500 ML-ready datasets²⁹. Integrated with human annotation, these supervised datasets cover a wide range of applications, such as land-use land-cover mapping³⁰, change detection³¹, object detection³², and disaster monitoring³³, as illustrated in Fig. 2b. The diverse array of applications along with semantic labels can steer the FMs toward nuanced understanding and broad applicability. Several recent works on FMs have also released a set of curated pre-training datasets without labels^{34–37}. Compared to the raw EO archive, these datasets are compact yet comprehensive, crafted with strategic sampling strategies that enhance the efficiency of model training. Nevertheless, in contrast to the EO data archive, existing labeled datasets in EO amount to less than 0.1 PB to date (see the right y-axis of Fig. 2a), i.e., only about 0.1% of the archive of open EO data, highlighting the huge discrepancy between them.

While the majority of historical Earth FM research has focused on EO and weather due to data availability, a significant volume of data is

Fig. 3 | The ideal Earth foundation model. The ideal Earth foundation model (FM) should possess eleven essential features. These features include geolocation embedding, balanced geographical representations, scale awareness, wavelength embedding, multisensory, the time variable, task-agnostic, carbon minimized, uncertainty quantification, physical consistency, and leveraging general-purpose vision–language models (VLMs)/large language models (LLMs). The first eight features (1–8) define the basic functionality of an ideal Earth FM, i.e., the ability to support any downstream task regardless of input data at any location in an environmentally friendly manner. The last three features (9–11) ensure the trustworthiness and human-centric design of an ideal Earth FM. (For attributions of figure elements, please see Supplementary Note 3).

Essential Features

1. Geolocation embedding
2. Balanced geographical representations
3. Scale awareness
4. Wavelength embedding
5. Multisensory
6. The time variable
7. Task-agnostic
8. Carbon minimized
9. Uncertainty quantification
10. Physical consistency
11. Leveraging general-purpose VLMs/LLMs



increasingly being used for Earth system applications, particularly in spatiotemporal modeling. Examples include FloodCastBench³⁸ for flood prediction, PDEBench³⁹ for scientific ML, a global community dataset⁴⁰ for large-sample hydrology, and soil moisture datasets⁴¹ for time series forecasting. These datasets, typically based on EO and in situ measurement data, are grounded in reliable physical models, forming a solid foundation for developing Earth surface and subsurface spatiotemporal FMs. Other sources such as natural language texts, cadastral records, and historical in-situ measurements are also valuable for the training of Earth FMs. These multimodal data sources, spanning diverse spatial and temporal scales, further underscore the central importance of developing a unified Earth FM capable of bridging scales.

Crafting the ideal Earth foundation model

An ideal Earth FM (as illustrated in Fig. 3) should take domain-specific characteristics (see Fig. 1) into account, and thus should feature:

1. **Geolocation embedding:** EO and climate data are linked to a geolocation. Embedding geolocation into the FM, either in a hard-coded way by importing the latitude and longitude as inputs, or in a smoother manner by contrasting features according to their geographical distances⁴², can improve feature representation or even facilitate the study of localized events, such as heatwaves or hurricanes. Researchers can potentially pinpoint the exact locations of environmental phenomena, enabling precise analysis of regional climate patterns and their global implications.
2. **Balanced geographical representations:** Ensuring that FMs reflect diverse geographical and ecological regions in their pre-trained weights is essential. Since models often over-represent data-rich areas like Europe and North America, achieving balanced geographical coverage requires careful selection of training data to reflect the diversity of Earth's surfaces, climates, and ecosystems, thus promoting fairness and reducing bias in downstream tasks. For example, without global coverage, important and unique biomes like rainforests and tundra may be missing from pre-training datasets, resulting in inaccurate estimates of biomass and carbon fluxes.
3. **Scale awareness:** The spatial resolution of EO and climate data varies from sub-meter for high-resolution aerial images to hundreds of kilometers for global climate models. The ideal FM should have scale awareness of different input data and learn spatial features referring to a certain characteristic scale. Practically, this is a mandatory feature for the model to deal with scale variances of the input data. More importantly, the FM should be capable of examining large-scale patterns, such as ocean circulation, while also delving into finer-scale processes, like land cover changes.
4. **Wavelength embedding:** Every individual satellite has a unique specification in terms of wavelengths and number of bands. The ideal FM

should incorporate data across different wavelengths, including visible light, infrared, ultraviolet, and microwave, and possess different numbers of channels, ranging from one (e.g., panchromatic) to several hundred (e.g., hyperspectral). Fine-tuning pre-trained models should be possible with input data of any wavelength and number of bands, including completely new and unseen sensors.

5. **Multisensory:** Multi-source and multisensory EO and climate data offer complementary information in order to serve various tasks. The FM should be capable of learning general feature representations across all data sources while retaining the unique characteristics of each source. This dual capability ensures that the model can draw upon a broad range of data and explore their joint and source-specific features to build a comprehensive picture of the Earth and its climate system. Meanwhile, preserving source-specific features allows for, e.g., the accurate attribution of climate phenomena to their respective forcings.
6. **The time variable:** EO and climate time-series data offer valuable new capabilities in monitoring the changing planet and making predictions with unprecedented spatial and temporal resolution. Therefore, the ideal FM should be able to learn sequential relations of data within its spatial context. This feature is vital, e.g., for distinguishing between natural variability and anthropogenic impact on the Earth surface, detecting anomalies, and identifying emerging patterns.
7. **Task-agnostic:** True FMs should be task-agnostic and exhibit strong generalization and transferability across diverse downstream tasks and domains. In EO, a single pre-trained encoder backbone should seamlessly adapt to image classification, regression, semantic segmentation, and object detection tasks across various applications, such as land use change detection, extreme event monitoring, or carbon cycle modeling. In weather and climate, although the model may be trained primarily to reproduce reanalysis data, it should generalize effectively to novel tasks and time periods, demonstrating the ability to predict future atmospheric phenomena for a range of downstream applications, including tropical cyclone tracking, atmospheric river forecasting, and extreme temperature prediction.
8. **Carbon minimized:** Given the urgency of addressing climate change, it is indispensable that the development and operation of FMs minimize their carbon footprint. This can be achieved through unified models for more general purposes, efficient fine-tuning⁴³, the use of renewable energy sources, and the distillation of model architectures using student–teacher networks. By prioritizing sustainability in its design, the ideal FM aligns with the broader goals of reducing greenhouse gas emissions and promoting environmental stewardship.
9. **Uncertainty quantification:** Quantifying uncertainty and detecting out-of-distribution instances, e.g., in their learned representations (embeddings), is a crucial ability for FMs, especially when adapted to extreme Earth events. In this situation, data scarcity and unexpected

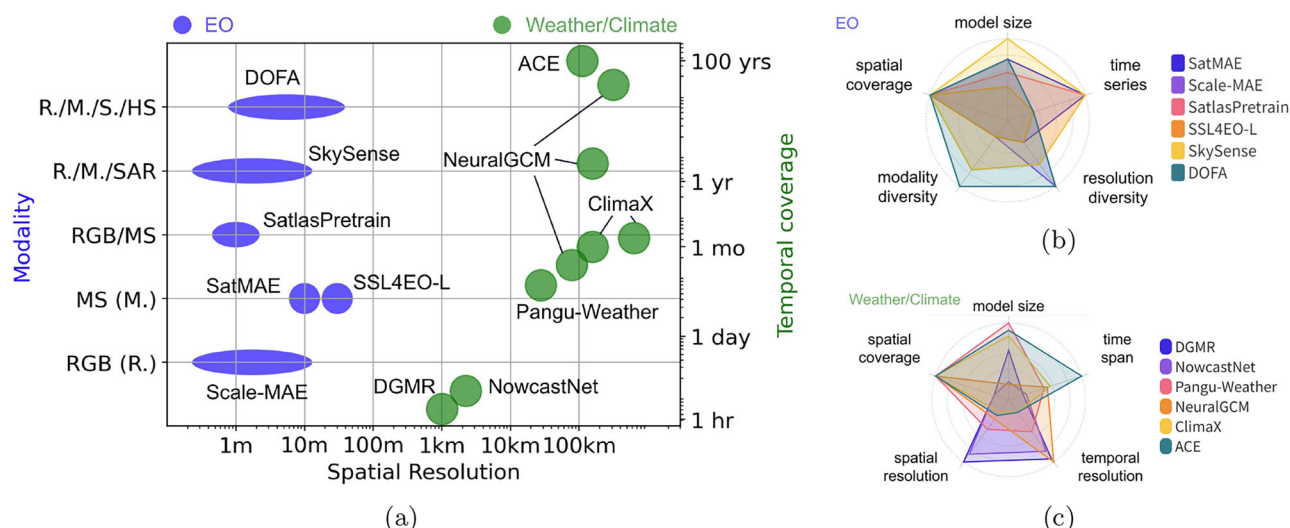


Fig. 4 | Representative Earth observation, weather, and climate foundation models. Representative models are chosen based on both popularity and novelty. **a** Comparison between Earth observation (EO), weather, and climate foundation models (FMs). Ellipses represent models trained on data with a range of spatial

resolutions. Modalities include red-green-blue (RGB), multispectral (MS), hyper-spectral (HS), and synthetic-aperture radar (SAR). **b** EO FM comparison. **c** Weather/climate FM comparison. No single model excels at all evaluation criteria, as most models focus on only one or two features.

situations can significantly impact the performance and robustness of downstream tasks. Considering this, enabling FMs with uncertainty quantification and out-of-distribution detection abilities can make them more reliable in critical applications, such as disaster response.

10. **Physical consistency:** Purely data-driven FMs can make predictions based on non-physical relationships or spurious correlations¹⁶. Incorporating principles of physical consistency, such as conservation, symmetry, and causality, improves the pre-training and fine-tuning of FMs, enhancing their transferability to new domains and increasing their transparency. This is especially important in the climate domain, where the ability to gauge the response of the model to different socioeconomic scenarios, including ones not seen at training time, is critical. Physically consistent models, alternatively termed knowledge-guided or physics-aware ML^{44,45}, represent a promising research direction to address the physical consistency of Earth FMs.
11. **Leveraging general-purpose VLMs/LLMs:** Developing Earth FMs that interpret EO, weather, and climate data with the support of LLMs enables complex reasoning, knowledge embedding, and interactive conversation. By combining embedded scientific knowledge with long-term memory and mechanisms for modeling system evolution over time, such models can simulate past, present, and potential future states of the Earth system. This reasoning capability also enables scenario analysis and supports causal inference, allowing users to explore counterfactual outcomes under alternative conditions. Such capabilities enhance not only the interpretability of Earth FMs but also their utility in decision support and hypothesis exploration. Moreover, natural language interfaces make these models more accessible and usable for both domain experts and the broader public.

Recent advances and gaps

Pioneering studies have been conducted on the development of FMs for EO (remote sensing), weather forecasting (instantaneous atmospheric states), and climate modeling (statistical distributions of weather over long time periods). However, compared to the ideal FM we describe above, existing models are still in their infancy, focusing on a few of the above essential features but rarely covering all eleven. In this section, we discuss the state of the art and current limitations with a focus on EO, weather, and climate FMs, which are typically developed separately and with different downstream applications in mind. A general view of several representative EO

and climate models is shown in Fig. 4. Figure 4a compares the spatial resolutions at which EO, weather, and climate models operate, while Fig. 4b and c present comparisons between several state-of-the-art models. As it stands, there exists a clear gap in the spatial resolutions at which EO, weather, and climate models operate. In order to create a unified Earth and climate model, these spatial and temporal scales must be linked.

Earth observation foundation models

EO has a relatively rich record of FM development, owing in part to the many similarities between remotely-sensed and natural images and the abundance of unlabeled data (Fig. 2a). Existing architectures, self-supervised techniques, and data augmentation methods can be conveniently adopted, leading to the widespread availability of EO FMs⁴⁶. Similar to computer vision models, EO models demonstrate *task-agnostic* representation capabilities when trained on million image-scale datasets^{37,47}. Nevertheless, the unique characteristics of EO data and applications call for domain-specific considerations⁴⁸, leading to many exciting recent advances.

SkySense⁴⁷ incorporates *geolocation* data directly into the model, embedding geolocation-related knowledge at the input level. SSL4EO^{35,36} addresses *geographical imbalance* by pre-training on datasets sampled from regions with high spatial variability, ensuring robust generalization across diverse geographical areas. Meanwhile, SatlasPretrain³⁷ focuses on scale, utilizing imagery that covers 21 million km², about 10% of the Earth's land surface. Scale-MAE tackles *scale awareness* by learning relationships between data at varying, known scales during pretraining.

For *wavelength embedding*, DOFA⁴⁹ employs a dynamic weight generation strategy based on wavelength, allowing it to adapt to unseen spectral signals. Similarly, SatMAE⁵⁰ groups subsets of spectral bands and integrates spectral encoding into positional encoding to capture wavelength information. SatMAE⁵⁰, SatlasPretrain³⁷, and SkySense⁴⁷ also integrate *time variables* into positional encoding to model temporal dynamics, crucial for tracking environmental changes.

As shown in Fig. 4a, many models support *multisensory* input to handle diverse data modalities. DOFA⁴⁹, for example, supports the widest range of sensors, including synthetic-aperture radar (SAR), red-green-blue (RGB), multispectral (MS), and hyperspectral (HS). Other models, such as SatMAE and SSL4EO-L, focus primarily on MS data from Sentinel-2 and Landsat. Training large-scale FMs typically consumes significant computational resources, yet few models currently prioritize *carbon minimization*. In this regard, GFM⁵¹ and DOFA⁴⁹ conduct continual pretraining to build

efficient FMs. The latter, as well as OFA-Net⁵², also aims to reduce energy consumption by training a unified FM across multiple modalities, avoiding the need to train separate models. Given their general-purpose design, all these models leverage self-supervised pre-training methods to support *task-agnostic* capabilities.

However, features like *uncertainty* and *physical consistency* remain areas of active exploration for EO FMs. When it comes to *language alignment*, the community has made progress incorporating language into vision modeling^{53–56} through approaches such as deriving text from semantic annotations^{53,57–59}, or using advanced vision–language FMs like GPT4-V⁶⁰ for captioning^{61,62}. However, these methods often oversimplify semantic information, generate irrelevant text, or are computationally expensive. Integrating textual data into EO remains largely underexplored as of 2024. Readers can refer to Supplementary Note 2 for a complete overview of vision–language datasets (Supplementary Fig. 1) and models in remote sensing.

Weather and climate foundation models

Compared to EO, the use of ML for weather forecasting and climate modeling is relatively recent. The question of whether or not recent advancements in deep learning could potentially contribute to or replace computationally expensive weather and climate models was first posed by ECMWF as early as 2018⁶³, and purely data-driven weather forecasting models did not begin to outperform traditional numerical models until 2022⁶⁴.

In this time, FMs have made great strides in *minimizing the carbon footprint* of weather forecasting. As an example, FourCastNet¹² is capable of generating a week-long global forecast in less than two seconds on a single GPU, requiring 12,000 times less energy than ECMWF's IFS⁶⁵. Faster prediction times also enable larger ensembles, bolstering capabilities for *uncertainty quantification*. However, it is worth noting that the majority of weather FMs are pre-trained on reanalysis products like ERA5, which itself required years of compute time on multiple high-performance computing clusters to produce²³. Aardvark Weather⁶⁶ represents a step towards reducing reliance on reanalysis products, with the ability to make global gridded forecasts and local station forecasts based solely on radar imagery and in situ station observations.

Architectural development for weather and climate models has progressed similar to that of EO models, with early works primarily based on simple convolutional neural networks (DLWP⁶⁷, MetNet⁶⁸). Vision Transformers (Pangu-Weather⁶⁴, ClimaX⁶⁹, FengWu⁷⁰, W-MAE⁷¹, and FuXi⁷²) now dominate the current landscape, using attention mechanisms that allow for better modeling of long-distance teleconnections. KeislerNet⁷³ first introduced the idea of using graph neural networks to handle latitudinal resolution changes, while GraphCast⁷⁴ successfully scaled up this approach from 1° to 0.25° resolution. More recently, generative models have seen growing popularity, including DGMR⁷⁵, which uses a generative adversarial network. Diffusion models including SEEDS⁷⁶ and GenCast⁷⁷ show promising results as well, with state-of-the-art performance on a number of tasks. Supplementary Table 2 summarizes the model architectures and features of many of these models, and Siddiqui et al.⁷⁸ provide detailed ablation studies of the model components that influence model performance.

The majority of modern weather and climate FMs are trained on *multisensor* reanalysis products like ERA5 at a global scale, and thus innately offer *geolocation embedding* and *balanced geographical representations*. Although ERA5 limits many models to a spatial resolution of 0.25°, a notable exception is the recent FengWu-GHR⁷⁹ model, which brings the spatial resolution down to 0.09° (about 10 km) using priors from pre-trained high-resolution models.

Weather and climate forecasting also happen at a number of *temporal* scales, including nowcasting (0–2 h), medium-range forecasting (2 h–2 weeks), subseasonal (2 weeks–2 months) to seasonal (2 months–2 years) forecasting, and long-term climate modeling (decadal scale). Nowcasting models tend to focus on regional scales, including MetNet⁶⁸ (USA)

and DGMR⁷⁵ (UK, USA), and show the most promise for integrating EO data into purely data-driven predictions. The majority of weather FMs fall under “medium-range forecasting”, and predict a number of *task-agnostic* atmospheric and surface phenomena every 6 h. The first weather FM to outperform traditional numerical models, Pangu-Weather⁶⁴, combines multi-resolution models to allow for predictions every 1 h. Subseasonal to seasonal (S2S) models like ClimODE⁸⁰ and Ola⁸¹, and long-term climate models like ACE⁸² and DLESyM⁸³ are less common, and even fewer models like NeuralGCM⁸⁴ have been tested at multiple different temporal scales.

As these models move from short-term prediction to long-term modeling, there is a gradual transition between predicting the instantaneous state of the atmosphere and learning the statistical distribution of weather states. The latter in particular lends itself well to a hybrid ML- and physics-based modeling approach, allowing certain guarantees of *physical consistency*. ACE⁸² and Ola⁸¹ are based on spherical Fourier neural operator⁸⁵ architectures, with ACE focusing on the conservation of mass and moisture and Ola focusing on coupled atmosphere–ocean modeling. NeuralGCM⁸⁴ represents a unique and promising modeling direction, with a fully differentiable general circulation model encompassing many physics-based atmospheric dynamics and outperforming the atmosphere-only physics-based AMIP⁸⁶ model. More recently, ClimODE⁸⁰ uses a physics-informed neural solver to model a global advection equation, allowing it to predict both weather and subseasonal climate.

The rapid progress of LLMs has also impacted climate models, making them more user-friendly and widely applicable through natural language integration. For instance, ClimateBert⁸⁷ facilitates climate text analysis, and ClimateGPT⁸⁸ serves as an *AI assistant* for climate change and sustainability topics.

Other Earth foundation models

Apart from EO and climate, there have been recent propositions involving the integration of FMs in the development of general geoscientific fields⁸⁹. In hydrology, HydroPML⁴⁴ introduces a physics-aware FM for flood and runoff forecasting, ensuring *physical consistency* in the water cycle. Generative ML techniques, such as diffusion models⁹⁰, are used to quantify *uncertainty* in spatiotemporal Earth system processes. Earth AI⁹¹ supports multiple tasks related to population and environmental dynamics, fostering *task-agnostic* Earth FMs. OceanGPT⁹², the first LLM for ocean science, demonstrates expertise across various ocean applications. In disaster management, DisasterResponseGPT⁹³ and DisasterM3⁹⁴ offer a language interface for tackling diverse disaster scenarios, acting as an *AI assistant* for disaster response. For scientific ML, Subramanian et al.⁸⁹ review the scaling and transfer properties of neural operators across partial differential equation systems. Additionally, FMs are employed for forecasting *temporal* dynamics, including variables such as wind⁹⁵ and water temperature⁹⁶.

Gaps towards the ideal model

Despite significant advances towards many of the features of an ideal FM, large gaps still remain. Works like SeCo³⁴ and SSL4EO that focus on pre-training dataset curation tend to oversample from densely-populated regions like cities and undersample from underpopulated regions like rainforests, polar regions, and oceans, each critical for ecology, climatology, and oceanography. Furthermore, pre-training datasets tend to focus on medium- to high-resolution RGB and MS, with few large-scale datasets for SAR, HS, or Lidar data or for very high- or low- resolution imagery⁴⁶. Multisensor models like DOFA are rare, and most models need to be trained from scratch on every new imaging platform encountered, making it difficult to take advantage of the thousands of satellites in orbit. Time-series EO modeling is also in its infancy, with few models able to handle variable length image sequences with irregular spacing. Integrating uncertainty quantification and physical consistency in EO remains underexplored.

Weather and climate FMs have come a long way in a short period of time, yet still show similar gaps in many aspects. Although inference is several orders of magnitude faster than traditional numerical models, training can be a bottleneck for many researchers. For example,

NeuralGCM⁸⁴ required 3 weeks to train on 256 TPU v4, while ORBIT⁹⁷ (weighing in at 113 billion parameters), was trained on a staggering 49,152 MI250X GPUs. Most FMs are currently limited by the 0.25° resolution of ERA5, and struggle to approach higher resolutions. Nowcasting and medium-range forecasting have seen great progress, while S2S forecasting and climate projection remain underexplored. Weather FMs suffers from overly smooth forecasts and bias drift as lead time increases⁹⁸, making long-range forecasting an active challenge. Climate FMs incorporating physics-informed ML have shown stability over long timescales, but often lack a closed energy balance, resulting in long-term drift and instability. Further work is needed to better model carbon, hydrologic, and biogeochemical cycles, where terrestrial surfaces and ocean currents play a big role.

As seen in Fig. 4a, there remains a large gap in spatiotemporal resolution/scale between the EO, weather, and climate domains, despite obvious connections between surface processes and weather. Table 1 illustrates current progress towards the ideal FM with several prominent examples.

The way forward

To achieve the ideal Earth FM, two factors must be carefully considered: (1) a comprehensive data curation process to ensure high-quality pretraining data, and (2) a model architecture with key features tailored to the specific challenges of these domains. Figure 5 presents a general workflow of different modules, which we dig deeper into below.

Data curation

- a. **Comprehensive data sources:** Data diversity is the foundation of an effective large-scale pretraining dataset. Satellite imagery, reanalysis products, and climate simulations are all important sources of data for model training. The diverse data sources ensure a holistic representation of the Earth system, and capture phenomena across scales and conditions. To ensure geographical diversity, it is vital to include data across evenly distributed geographical regions to avoid data bias. To improve modality diversity, different data modalities including language data are crucial to capturing complex data relationships and dynamic processes. In practice, applying the FAIR (Findable, Accessible, Interoperable, Reusable) data principles to these sources helps ensure that heterogeneous geospatial datasets can be consistently pulled into FM pretraining pipelines.
- b. **Metadata integration and standardization:** Metadata is the crucial bridge linking diverse observation data sources, and plays a key role in organizing data observed from Earth's surface and atmosphere. Metadata including the data source, collection time, modality, resolution, cloud coverage, spectral bands, and preprocessing details are critical for enabling FMs with features like geolocation embedding, scale awareness, and wavelength embedding. As for weather and climate phenomena, it is important to employ standardized ontologies for each data point to maintain consistency across datasets.

Model design and training

- c. **Dynamic encoder with conditional computation:** A dynamic encoder leveraging conditional computation techniques enables the flexible integration of varying numbers of data modalities for fusion. Techniques such as autoregressive sequence modeling for multi-modal feature fusion and metadata-guided dynamic feature projection⁴⁹ offer the flexibility to adapt to diverse data modalities. Even when certain modalities are missing in downstream tasks, the model can still function effectively, leveraging complementary features from the available modalities. This design enhances the model's robustness in handling multi-modal, multi-scale data and supports advanced capabilities such as geolocation embedding, wavelength embedding, and scale awareness.
- d. **Alignment with large language models:** Merging EO, weather, and climate FMs with language processing capabilities is vital for enhancing

knowledge embedding, improving reasoning abilities via reinforcement learning, and facilitating more effective user interaction⁹⁹. By designing effective projection modules, Earth FMs can learn to align with language capabilities to gain open-vocabulary learning ability¹⁰⁰, and enrich the understanding of an extensive range of terms and concepts.

- e. **Unified FM with geographical mixture of experts:** Designing an effective Geographical Mixture of Experts¹⁰¹ (GeoMoE) strategy is pivotal to harnessing the collective strengths of FMs in the Earth sciences. GeoMoEs integrate specialized models across various domains, enabling the framework to precisely allocate the right expertise. GeoMoEs not only enhance performance by adapting to diverse situations but also reduce computational costs by judiciously deploying resources. The adaptable nature of GeoMoEs allows for the continuous integration of emerging scientific discoveries, ensuring the FMs remain at the forefront of technological and scientific progress.
- f. **Multi-modal learning framework:** An ideal FM, pretrained with data from multiple sources, must skillfully exploit the synergy among diverse data modalities, weaving together complementary insights and common knowledge¹⁰². To accomplish this, methods that concentrate on two essential aspects of multi-modal learning need to be designed^{103,104}. One is meta-modality representation learning, which extracts and synthesizes information across various modalities, creating a meta-representation that captures overarching patterns. The other is modality-specific representation learning, which utilizes modality-specific encoders to learn rich, modality-specific features.
- g. **Spatio-temporal analysis across multiple scales:** EO data is collected at various spatial and temporal resolutions, capturing processes at different scales. To effectively model these processes, scale-aware hierarchical spatiotemporal attention mechanisms have a high potential for establishing correlations between fine-scale Earth surface phenomena, larger-scale weather events, and global climate patterns. These mechanisms allow the model to understand how conditions evolve and interact across different regions and scales, thereby enhancing the model's performance in understanding Earth's dynamic systems. Furthermore, integrating data assimilation techniques into FMs¹⁰⁵ offers a principled way to fuse real-time or historical observational data with model-internal representations. This enables the models to maintain temporal coherence with the evolving Earth system, improving forecasting accuracy and aligning predictions with physical reality across multiple scales.
- h. **Self-/weakly-supervised learning:** To learn generic information from big EO and climate data, scalable self-/weakly-supervised pretraining techniques are necessary to be explored. Predictive, contrastive, and generative self-supervised learning methods pave the way to capture hierarchical representations for different downstream tasks⁴⁶. Weakly-supervised methods can help better utilize semantic information from existing coarse or noisy annotations.
- i. **Continual pre-training and distillation:** Incorporating continual learning mechanisms⁵¹, regularization techniques, and transfer learning strategies is critical to creating a carbon-minimized and adaptable Earth FM. These strategies allow the model to persistently update its knowledge, enhancing performance and generalization in response to new and evolving scenarios. Techniques like elastic weight consolidation¹⁰⁶ to prevent catastrophic forgetting and knowledge distillation for efficient model scaling are pivotal.
- j. **Uncertainty quantification:** In EO and climate modeling, when forecasting weather or natural disasters, efficient uncertainty quantification is critical. Sparse Gaussian processes¹⁰⁷ offer a scalable Bayesian approach, using inducing points to manage the computational load. Ensemble methods¹⁰⁸ utilize multiple models to capture a broad spectrum of outcomes, enhancing performance and providing robust uncertainty measures. Quantile regression¹⁰⁹ directly computes uncertainty intervals, efficiently offering insights into prediction reliability. These

Table 1 | Representative Earth observation, weather, and climate foundation models and which of the eleven features of an ideal foundation model have been addressed in their model design or training procedure

FMs	1. Geolocation Embedding	2. Balanced Distribution	3. Scale Awareness	4. Wavelength Embedding	5. Multisensory	6. The Time Variable	7. Task-agnostic	8. Carbon-minimized	9. Uncertainty	10. Physical Consistency	11. Language Alignment
SatMAE ⁵⁰	X	X	X	✓	X	✓	✓	X	X	X	X
Scale-MAE ¹³⁸	X	X	✓	X	X	X	✓	X	X	X	X
SatlasPretrain ³⁷	X	X	X	X	✓	✓	✓	X	X	X	X
SSL4EO-L ³⁶	X	✓	X	X	✓	X	✓	X	X	X	X
SkySense ⁴⁷	✓	X	X	X	✓	✓	✓	X	X	X	X
DOFA ⁴⁸	X	X	X	✓	✓	X	✓	✓	X	X	X
Copernicus-FM ¹³⁹	✓	✓	✓	✓	✓	✓	✓	✓	X	X	X
TerraMind ¹⁴⁰	✓	✓	X	X	✓	X	✓	X	X	X	✓
DOFA-CLIP ¹⁴¹	X	✓	X	✓	✓	X	✓	✓	X	X	✓
AlphaEarth ¹⁴²	✓	X	X	X	✓	✓	✓	X	X	X	✓
DGMR ⁷⁵	X	X	X	-	✓	✓	X	X	✓	X	X
NowcastNet ¹⁴³	X	X	✓	-	✓	✓	X	X	X	✓	X
Pangu-Weather ²⁴	✓	✓	X	-	✓	✓	✓	X	✓	X	X
NeuralGCM ⁸⁴	✓	✓	✓	-	✓	✓	✓	X	✓	✓	X
Aurora ¹¹⁵	✓	✓	✓	-	✓	✓	✓	X	X	X	X
ClimaX ⁴⁹	✓	✓	X	-	✓	✓	✓	X	X	X	X
ACE ⁸²	✓	✓	X	-	✓	✓	X	✓	✓	✓	X

Wavelength embedding is ignored for climate foundation models (FMs) due to the lack of a wavelength component in atmospheric measurement data. Note that many climate FMs are carbon-minimized at inference time (compared to numerical weather prediction models), but still require substantial computing resources during training. Also note that while this table is presented as binary checkboxes, all features are in fact scales, and much work remains to improve the spatial, spectral, and temporal resolutions of these models.

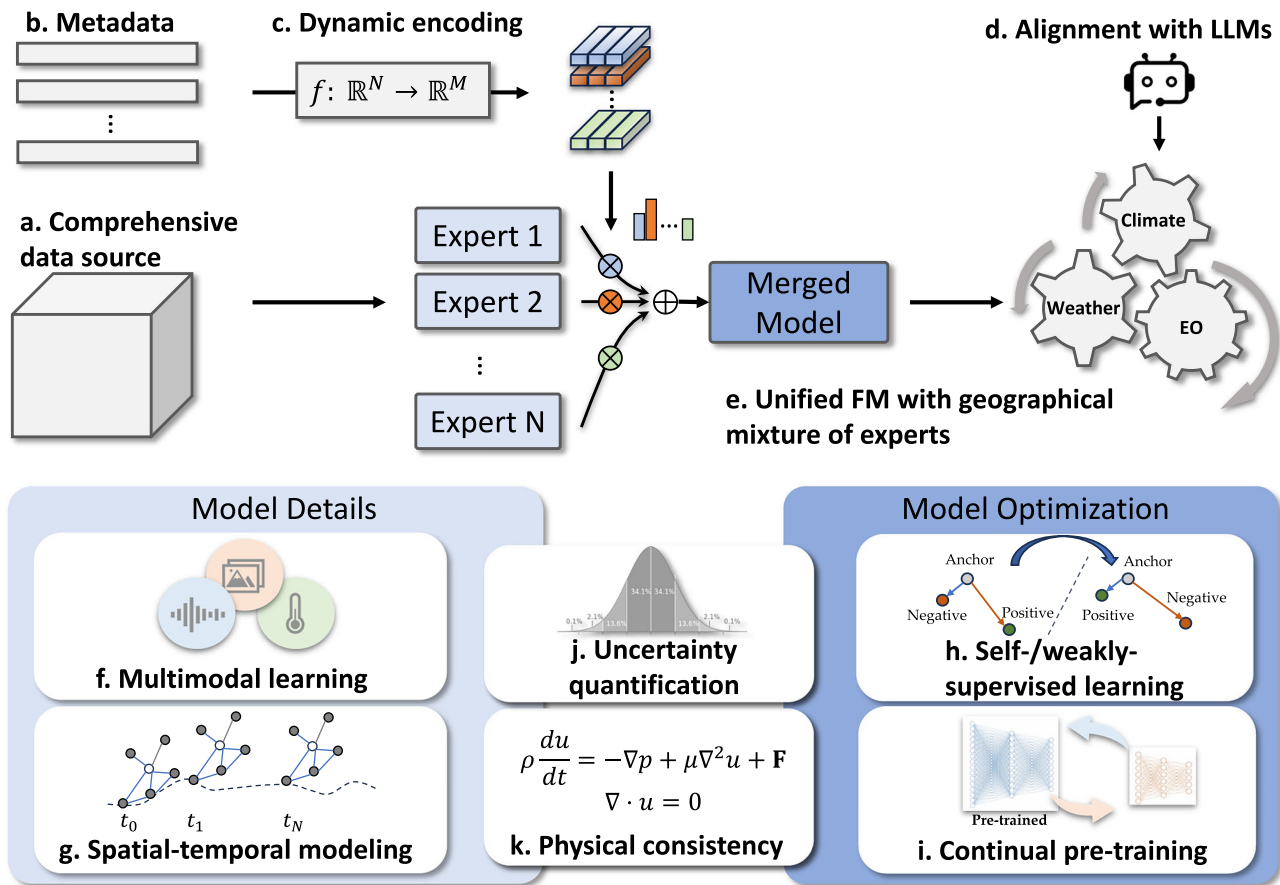


Fig. 5 | The way forward towards the ideal Earth foundation model. The way forward spans from data curation (a, b) to model design and training (c–k). The top row includes an overall schematic, while the bottom row includes specific details of

model implementation (f, g) and optimization (h, i), and components that could involve both (j, k). Large language models (LLMs) may be required to align Earth observation (EO), weather, and climate foundation models (FMs).

candidate techniques have the potential to offer FMs the uncertainty quantification abilities required in the Earth sciences.

- k. **Physical consistency:** A number of physical principles must be respected to accurately model and predict natural phenomena, including known physical equations (such as dynamical equations), physical properties (such as conservation, symmetries, and causality¹¹⁰), and physical conditions (such as initial and boundary conditions). Physical consistency can be achieved through the integration of physics into datasets, architectures, or loss functions, thereby augmenting the FM's generalization and interpretability. For instance, physical knowledge is usually represented by differential equations with analytical or numerical solutions. Physics-informed neural networks^{111,112} incorporate these equations into the neural network's loss function, enabling unsupervised learning for understanding physical processes. Physics-informed neural solvers⁸⁰ provide a framework for solving the continuity equation globally by formulating it as a system of neural ordinary differential equations. In addition, physics-embedded neural networks¹¹³ can incorporate the intrinsic preservation of physical properties into their architecture to enhance the efficiency of learning underlying physical phenomena. Incorporating knowledge of climate processes¹¹⁴, particularly their invariance, into data-driven models of Earth system processes can significantly enhance the models' consistency and data efficiency across various climate regimes. Rapid climate emulators⁸² enable the assessment of

fundamental physical laws, such as the conservation of mass and moisture, by integrating diagnostic variables into the model. Leveraging diverse geophysical data and domain-specific knowledge in geoscience, we can establish a large-scale FM for the Earth system¹¹⁵. Furthermore, employing a hybrid learning framework⁸⁴ that integrates physical models and ML within a differentiable neural network can result in the establishment of a structured geodynamics FM.

Evaluating Earth foundation models

Task-specific models are usually evaluated against a specific test dataset. Summarizing model performance through numerical metrics such as accuracy or F1-score allows for easy comparison of such models. FMs, however, are supposed to be beneficial for a wide range of tasks, posing challenges to the comparability of FMs. In natural language processing, FMs are evaluated through so-called *benchmarks*, which specify a collection of evaluation tasks. The tasks within a benchmark are selected to evaluate different aspects of downstream applications¹¹⁶. In computer vision, benchmarks are mostly assembled ad hoc from existing evaluation datasets¹¹⁷.

Useful Earth FMs must learn robust representations for diverse EO, weather, and climate data. Evaluating these models requires standardized benchmarking suites similar to the ones proposed in natural language processing to assess model performance across various tasks and datasets. This section discusses key requirements for such benchmarks and examines existing benchmarks, highlighting their strengths and limitations. Ultimately, we identify gaps where future benchmarks can offer even more comprehensive evaluations of Earth FMs.

Requirements for Earth benchmarks

- **Data Diversity:** Benchmarks for Earth FMs need to evaluate the model on diverse data. For EO, various data modalities need to be covered, such as multispectral, SAR, and hyperspectral data. For weather and climate data, various variables should be included, like atmospheric pressure, temperature, and precipitation. Another aspect of data diversity is spatio-temporal coverage of the evaluation tasks, which should represent most parts of the globe and range from near real-time applications to long time-series, including historical data. For climate data, even future projections should be included.
- **Representative Tasks:** Furthermore, the tasks in the benchmark need to be representative of downstream tasks that the models could be applied to. For EO, this includes tasks like scene classification, object detection, segmentation, change detection, time-series analysis, and regression. For the weather and climate domains, possible tasks are forecasting, extreme event detection, downscaling, or emulation of physical climate models and corresponding climate scenario analysis.
- **Interaction between components:** Ideally, a benchmark for Earth FMs should not only evaluate the model on its individual aspects of EO, weather, and climate, but also evaluate their integration. For this, tailored tasks should be defined that require the interaction between components. Possible examples for such tasks are downscaling weather data using satellite imagery, combining weather and satellite data to detect extreme events like droughts, or improving long-term climate projections with knowledge about short-term weather patterns.
- **Physical consistency:** While EO lends itself well to simple accuracy metrics, weather and climate require additional validation to ensure consistency with the physical world. Simply minimizing the root mean square error and anomaly correlation coefficient can result in overly smooth forecasts⁹⁸. S2S forecasts should exhibit known teleconnections, including the El Niño–Southern Oscillation and the North Atlantic Oscillation. Long-term climate benchmarks must ensure conservation of mass and energy, especially as it pertains to the carbon and hydrologic cycles. Additionally, benchmarks should assess causal reasoning and sensitivity to ensure that FMs' predictions react appropriately to controlled perturbations and that key drivers are correctly identified, enhancing robustness and interpretability.

Existing benchmark datasets

For evaluating EO models, SustainBench¹¹⁸ focuses on 7 of the United Nations' sustainable development goals, encompassing 15 tasks. The tasks are representative, covering classification, segmentation, regression and time-series analysis. It evaluates on a number of optical sensors like Landsat, Sentinel-2 and MODIS. However, SAR data is not considered in this benchmark. Finally, owing to the design of the dataset as a sustainability benchmark, the considered tasks are limited to agricultural and humanitarian applications. More recently, GEO-Bench¹¹⁹ is a benchmarking suite designed specifically for evaluating EO FMs. It includes diverse Earth-relevant tasks and has standardized evaluation protocols. By limiting evaluation to classification and segmentation tasks, however, it omits important EO tasks, such as regression or time-series analysis. MDAS¹²⁰ and PhilEOBench¹²¹ are suites that trade larger datasets for a smaller number of tasks. Finally, Prithvi¹²² and SkySense⁴⁷ have proposed new benchmarks alongside their FMs. Table 2a summarizes these benchmarks by the evaluated modalities, tasks, resolutions, and timespan.

For weather data, the WeatherBench¹²³ and WeatherBench 2¹²⁴ collections provide a comprehensive benchmark for medium-range weather forecasts on a global scale. These datasets include both historical records and model forecasts of essential meteorological variables, allowing for standardized evaluation of weather FMs. An increasing number of weather FMs are directly evaluated on ERA5^{64,72,77}. ExtremeWeather¹²⁵ specifically evaluates weather models on extreme weather events. Nowcasting is typically done only for precipitation⁶⁸, for which highly resolved datasets such as MRMS¹²⁶ exist. As a first multi-

task benchmark, ClimateLearn¹²⁷ evaluates not only weather forecasting, but also extreme weather prediction and downscaling.

ClimateBench¹²⁸ collects physical simulations under different forcing scenarios to aid the development and evaluation of data-driven climate emulation models. Similar to the trend of evaluating weather models directly on ERA5, some climate models are also directly evaluated against CMIP6 simulation outputs²⁷. Weather and climate benchmarks are summarized by evaluation tasks, resolution and timespan in Table 2b.

Gaps in existing benchmarks

Currently, there is *no consensus on how to adapt a pre-trained FM for a downstream task*. Some works fine-tune the entire model, updating all parameters. Other works only train a single final network layer in a process called linear probing. Others use classical ML models like k -NNs instead of linear layers. Finally, parameter-efficient fine-tuning methods are another option⁴³. Benchmark results will only be comparable when models have been adapted to the downstream tasks using the same fine-tuning method. In particular, benchmarks are missing for few- and zero-shot settings, where the model needs to generalize to new tasks with only a few labeled samples or without any labels.

However, *certain regions are not represented* in existing EO benchmarks. This limits the potential for cross-disciplinary research and holistic understanding of global climate systems. Including benchmarks for the missing regions such as the oceans¹²⁹ or polar areas¹³⁰ in evaluations would facilitate better modeling of phenomena like sea-level rise or ocean circulation patterns.

Further, evaluation datasets are often filtered for specific acquisition conditions like minimal cloud cover, which is *not representative of real-world data*. This can introduce artificial biases in the evaluation scores, making the models appear more capable than they truly are when applied to the diverse and often imperfect data encountered in actual use. Similarly, weather and climate benchmarks are often based on reanalysis products and might benefit from evaluations on observational data as well.

Especially for long-term climate modeling, the complex target distributions of climate variables imply that *summarizing evaluations with numerical metrics is not always possible*. For example, a climate predictor may have comparably good RMSE values when evaluated against climate simulations, but not capture the underlying processes well. Instead, works on weather and climate FMs often assess whether the models can capture specific climate scenarios, such as tropical warming trends⁸⁴, atmospheric rivers⁷⁴, or the tracks of tropical cyclones⁷⁴. While such scenarios are hard to evaluate with numerical metrics, a standardization of such scenarios is also needed to allow for inter-comparability between different weather and climate FMs.

For evaluating Earth FMs, the main gap in existing benchmarks is *the lack of evaluation tasks that integrate EO data with weather and climate data*. This limits our ability to assess how well these models can capture the complex interplay between land, ocean, atmosphere, and human activity.

The rapid development of Earth FMs currently faces a challenge with *the lack of standardized evaluation benchmarks*. With each model employing its own unique set of tasks, objective comparisons and assessments are difficult. A convergence towards standardized models is needed to ensure fair and transparent evaluations, and allow researchers to determine the strengths and weaknesses of each model. In particular, the development of such benchmarks should actively engage end users, whose domain expertise and operational needs are essential for defining evaluation criteria that remain unbiased and relevant in real-world settings. This, in turn, would greatly aid collaboration and accelerate the advancement of FMs for the Earth sciences.

Perspectives

What comes after the ideal Earth FM? Identifying the most relevant research questions is crucial for maximizing the utility of the model in advancing our understanding of Earth and climate science. We give seven recommendations, as follows:

Table 2 | Overview of existing benchmark datasets for Earth observation and weather/climate foundation models

(a) Benchmarks for Earth observation foundation models.													
Name	# Tasks	RGB	Multispectral	Hyperspectral	SAR	Time Series	Classification	Regression	Segmentation	Object Det.	Change Det.	Resolution	Timespan
SustainBench ¹¹⁸	15	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	0.6–30 m	1996–2019
GEO-Bench ¹¹⁹	12	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	0.1–15 m	2001–2021
FoMo-Bench ¹⁴⁴	16	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	0.01–60 m	2011–2023
MDAS ²⁰	3	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	0.3–30 m	2018
PhilEO Bench ¹²¹	3	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	10 m	unknown
SkySense ⁴⁷	17	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	0.05–30 m	2002–2019
Prithvi ¹²²	4	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	30 m	2017–2022
PANGAEA ⁴⁵	11	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	0.8–30 m	2015–2022
(b) Benchmarks for weather and climate foundation models.													
Name	Nowcasting	Medium-Range	S2S Forecasting	Climate Projection	Extreme Events	Downscaling	Resolution (Lat.)	Resolution (Lon.)	Resolution (Time)	Resolution (Time)	Resolution (Time)	Timespan	
ERA5 ⁴²³	✓	✓	✓	✓	✓	✓	0.25°	0.25°	1 h	1 h	1 h	1959–2024	
WeatherBench ¹²³	✓	✓	✓	✓	✓	✓	5.63°	5.63°	6 h	6 h	6 h	1979–2018	
WeatherBench 2 ¹²⁴	✓	✓	✓	✓	✓	✓	1.5°	1.5°	6 h	6 h	6 h	1979–2023	
ExtremeWeather ¹²⁵	✓	✓	✓	✓	✓	✓	0.23°	0.31°	6 h	6 h	6 h	1979–2005	
ClimateLearn ¹²⁷	✓	✓	✓	✓	✓	✓	5.63°	5.63°	6 h	6 h	6 h	1979–2018	
CMIP6 ⁴²⁷	✓	✓	✓	✓	✓	✓	1.25°	2.5°	24 h	24 h	24 h	1850–2100	
ClimateBench ¹²⁸	✓	✓	✓	✓	✓	✓	1.89°	2.5°	1 year	1 year	1 year	1850–2100	
MRMS ⁴¹²⁶	✓	✓	✓	✓	✓	✓	0.01°	0.01°	2 min	2 min	2 min	2014–2024	

^aRaw dataset without associated evaluation protocol.

- **Earth embedding:** Upon the readiness of an ideal Earth FM, we should leverage the big Earth data to construct an Earth Embedding—a unified, interpretable, and reliable representation space learned through the ideal Earth FM. This embedding shall be compact yet semantically rich, enabling users to directly exploit pre-learned representations for diverse downstream applications without the need to access raw EO data, thereby democratizing the AI usage of the big Earth data and reducing the carbon footprint associated with it.
- **Energy-efficient adaptation:** Upon acquiring FMs, enhancing their efficiency for downstream tasks is vital for broad applications. Future research should explore strategies to boost computational, parameter, and memory efficiencies without compromising predictive performance. Possible research directions include quantization of FMs, pruning to eliminate redundant parameters of FMs for faster inference in specific downstream tasks, and knowledge distillation to transfer insights from larger to smaller models. These efforts aim not only to broaden the models' applicability but also to advance toward more sustainable and energy-efficient ML development, especially in edge devices.
- **Foundation model-enabled sciences:** Once an ideal Earth FM becomes available, the most prominent research question is EO-informed weather and climate forecasting. As the FM is jointly trained from EO, weather, and climate data, it can be explored to address the scale mismatch between observational data and the physical model of the Earth system and its climate. Exploiting feedback at higher spatial resolution can enable new exciting forecast and management applications. One additional example is to detect and predict extremes. As the FM is trained on massive amounts of data with “normal” patterns, it can flag deviations with much higher accuracy, helping detect “extremes”, such as wildfires, floods, or other environmental disturbances. Furthermore, effectively combining the transferability of FMs with the interpretability of physical models offers a promising avenue for addressing extreme-event prediction challenges, such as forecasting floods in ungauged basins or projecting climate extremes from scarce data.
- **Machine unlearning:** An ideal EO FM utilizes multimodal data, such as high-resolution and multispectral images, which can violate privacy by exposing sensitive information about individuals and infrastructures. Even basic optical imagery can pose a threat to privacy. Addressing this, the concept of “machine unlearning”, coined in 2015¹³¹, aims to enable these models to effectively “forget” specific subsets of a dataset while minimizing negative effects on downstream task performance. This is crucial for compliance with regulations, as models' generalization abilities partly rely on memorization¹³², posing a trade-off between data protection and model performance.
- **Continual learning:** Efficiently updating FMs is also important. This includes integrating data from new sensors or modalities, upgrading outdated old data with better-quality new data, and ensuring up-to-date knowledge or constraints of the model under a changing climate. A natural technique is continual learning, which aims to allow models to learn new information without forgetting the past. However, compared to continual learning on task-specific applications, continual learning with generic FMs is more challenging due to model complexity, data size, computational cost, and task-agnostic objectives. To solve these, it's crucial to develop new model architectures, training objectives, and more sophisticated optimization mechanisms.
- **Adversarial defenses:** Furthermore, the continuous learning data integration process should be complemented with incorporating adversarial ML techniques to mitigate risks on FMs. This approach trains models to resist adversarial data, which are deliberately manipulated inputs designed to mislead the FM into making incorrect predictions¹³³. For instance, adversarial inputs might include subtly modified images, despite being barely noticeable, which can lead to inaccuracies in predictions and affect performance. By integrating adversarial ML, FMs become more robust

and better equipped to adapt to new patterns when deployed in real-world scenarios.

- **Interpretability:** FMs with billions of parameters trained on massive unlabeled data present particular challenges in explaining their outputs, compared to smaller, specialized ML systems. The nature of FMs, i.e., serving as the backbone for a wide range of downstream applications, makes research on their interpretability of crucial importance. This can be achieved to a certain extent, e.g., by examining token prioritization, directly adjusting model weights, interpretable distillation, and LLM-based explanation techniques. Nevertheless, these tools may introduce biases due to inappropriate model setup, non-physical reasoning, and the absence of counterfactual reasoning capabilities and should therefore be used with caution.
- **Cross-disciplinary inspiration:** Concepts from image FMs could inspire future improvements for Earth FMs, such as new architectures based on state-space models¹³⁴ or new ways of training FMs, like label-free scene understanding¹³⁵. Following similar requirements for multimodal FMs, medical AI is currently moving towards combining highly heterogeneous sets of modalities within FMs¹³⁶. For text FMs, retrieval augmented generation¹³⁷ is currently opening up many new applications. Future Earth FMs could similarly retrieve data that is geospatially or temporally close to a given query to improve their responses.

Data availability

Code and data used to create the figures in this paper are available at <https://github.com/zhu-xlab/EarthFoundationModels>.

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Author contributions

X.X.Z. conceived the work, created an initial outline and main storyline, and contributed to all sections. Z.X. and Y.W. contributed several sections of the manuscript, notably those focusing on Earth observation and model design. A.J.S. contributed several sections of the manuscript, notably those focusing

on weather forecasting and climate modeling. K.H. contributed several sections of the manuscript, notably those focusing on benchmarking and evaluation. Y.Y.W. designed and incorporated several of the primary figures, Z.Y. and T.D. contributed sections related to vision–language models, Q.X. contributed several sections of the manuscript, notably those focusing on physical consistency, and Y.S. contributed to the evaluation section.

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Correspondence and requests for materials should be addressed to Xiao Xiang Zhu.

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